

Chapter 3: Proteins - Test Questions

Vocabulary Matching

Match the term with its correct definition.

1. Polypeptide backbone	a) A segment of a polypeptide chain that can fold independently into a compact, stable structure.
2. α -helix	b) The spontaneous refolding of a denatured protein into its original conformation.
3. β -sheet	c) A protein that assists in the folding of other proteins.
4. Protein domain	d) The repeating sequence of atoms along the core of a polypeptide chain.
5. Quaternary structure	e) A rigid, cylindrical folding pattern stabilized by internal hydrogen bonds.
6. Molecular chaperone	f) A rigid, pleated structure formed by hydrogen bonds between adjacent polypeptide strands.
7. Renaturation	g) The complete structure of a protein complex formed from multiple polypeptide subunits.
8. Active site	h) A regulatory site on a protein distinct from the active site.
9. Allosteric site	i) The region of an enzyme that binds to the substrate.
10. Coiled-coil	j) A stable structure formed when two or more α -helices wrap around each other.
11. Ligand	k) A small molecule that binds to a protein.
12. Enzyme	l) A protein that catalyzes a specific chemical reaction.
13. Disulfide bond	m) A covalent bond that cross-links parts of a polypeptide chain or different chains.
14. Prion	n) An infectious, misfolded protein that can cause disease.
15. Ubiquitylation	o) The process of attaching a small protein to a target protein to mark it for degradation or other functions.

Multiple Choice

1. Which of the following is NOT a noncovalent bond that helps proteins fold? a) Hydrogen bonds b) Peptide bonds c) Electrostatic attractions d) Van der Waals attractions

2. The primary structure of a protein refers to its: a) Three-dimensional shape b) The sequence of its amino acids c) The arrangement of its subunits d) The presence of α -helices and β -sheets
3. An α -helix is stabilized by hydrogen bonds between: a) Side chains of adjacent amino acids b) The N-H and C=O groups of the polypeptide backbone c) Two different polypeptide chains d) Polar side chains and water
4. What is the main driving force for protein folding in an aqueous environment? a) The formation of disulfide bonds b) The hydrophobic clustering of nonpolar side chains c) The formation of peptide bonds d) Electrostatic attraction between charged side chains
5. A protein domain is defined as: a) The entire polypeptide chain b) A small, flexible linker region c) A segment of polypeptide that can fold independently d) The complex of multiple protein subunits
6. Which level of protein structure describes the arrangement of multiple polypeptide subunits?